Assignment – Object Oriented Programming

Instructions:

All assignment programs should first be written by hand in a notebook, and then implemented and executed in Jupyter Notebook.

Please include proper comments in the code to make it easy to understand and readable.

Question 1.

Complete the following steps:

- Create a class called `Person`
- Add an `__init__` method that takes `name` and `age` as parameters
- Add a method called `greet` that prints **"Hello, my name is "** followed by the name
- Create an object `p1` of the class with name **"John"** and age **36**
- Call the `greet` method on `p1`

Solution: **(Sample Comment)**

```
# Create a class
        #write __init__ method


        # write greet method


# Create an object


# Call the greet method
```

Question 2.

Complete the following steps:

- Create a class called `Dog`
- Add an `__init__` method that takes `name` and `age`, and store them as properties using `self`
- Add a method called `bark` that prints the dog's name followed by **" says Woof!"**
- Create an object `d1` of the `Dog` class with name **"Buddy"** and age **3**
- Call the `bark` method on `d1`

Question 3.

Complete the following steps:

- Create a class called `Car`
- Add an `__init__` method with a `brand` parameter, and store it as a property
- Add a method called `show` that prints the brand
- Create an object `c1` of the `Car` class with with brand **"Ford"**
- Call the `show` method on `c1`

Question 4.

Complete the following steps:

- Create a class `Student` with an `__init__` that takes `name` and `grade`, and stores them as properties
- Create an object `s1` with name **"Anna"** and grade **"A"**
- Print the grade of `s1`
- Change the grade of `s1` to **"B"**
- Print the updated grade

Question 5.

Complete the following steps:

- Create a class called `Rectangle`
- Add an `__init__` method with `width` and `height`, and store them as properties
- Add a method called `area` that returns the width multiplied by the height
- Create an object `r1` with width **5** and height **3**
- Print the area of `r1`

Question 6.

Complete the following steps:

- Create a parent class `Animal` with an `__init__` that takes `name`
- Add a method `speak` that prints the name
- Create a child class `Dog` that inherits from `Animal`
- Create an object `d1 = Dog("Rex")`
- Call `d1.speak()`

Question 7.

Complete the following steps:

- Create a class `Cat` with a method `sound` that prints **"Meow"**
- Create a class `Fox` with a method `sound` that prints **"Wa-pa-pa-pa-pa-pow!"**
- Create objects `c1 = Cat()` and `f1 = Fox()`
- Call `sound()` on both objects using tuple

Question 8.

Complete the following steps:

- Create a class `ScoreBoard`
- Add an `__init__` with a `score` parameter and store it as a private attribute
- Add a method called `get_score` that returns the private score
- Create an object `s1` with a score of **0**
- Print the score of `s1`